

Scramble

A FULLY AUTOMATED RUBIK'S CUBE SOLVER

PRESENTED BY -

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MOTIVATION

PROBLEM STATEMENT

DESIGN AND DEVELOP AN **AUTOMATED RUBIK'S CUBE SOLVER** USING A MICROCONTROLLER-BASED SYSTEM THAT CAN **PHYSICALLY MANIPULATE THE CUBE** AND **TRACK SOLVE TIME**, WITH **USER INPUT PROVIDED VIA A MOBILE INTERFACE**

SIGNIFICANCE

- The Rubik's Cube is a globally recognized puzzle that represents a challenge in both logic and dexterity.
- Automating its solution not only demonstrates **practical applications of embedded systems, robotics, and computer vision**, but also showcases how mechanical and software components can work together to solve complex problems.
- This project bridges the gap between theoretical concepts and real-world implementation, making it an excellent learning opportunity in the fields of mechatronics and embedded systems.

PERSONAL INTEREST

All three of us were fascinated by the Rubik's Cube back in **7th grade**, when we each owned multiple and took the time to learn how to solve it.

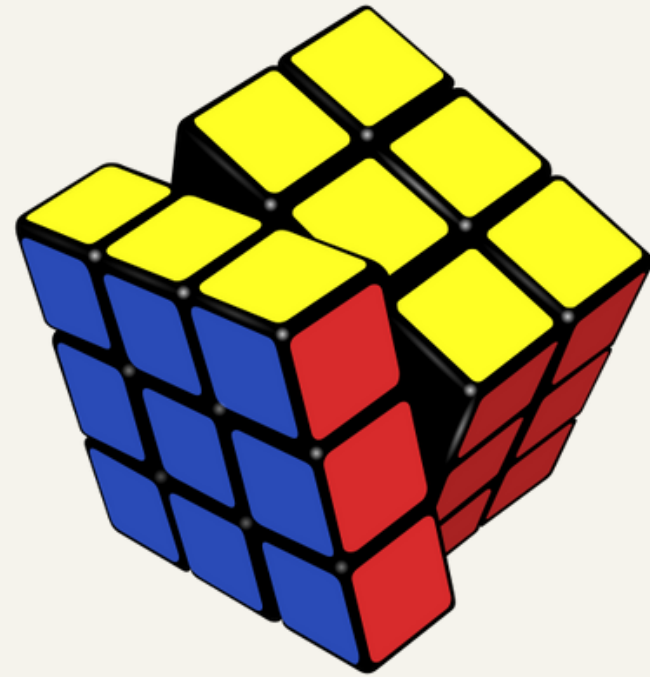
That early interest sparked our curiosity in problem-solving and logical thinking, and eventually inspired this project. This project is especially interesting to us as it showcases how **a childhood curiosity can grow into an engineering challenge.**



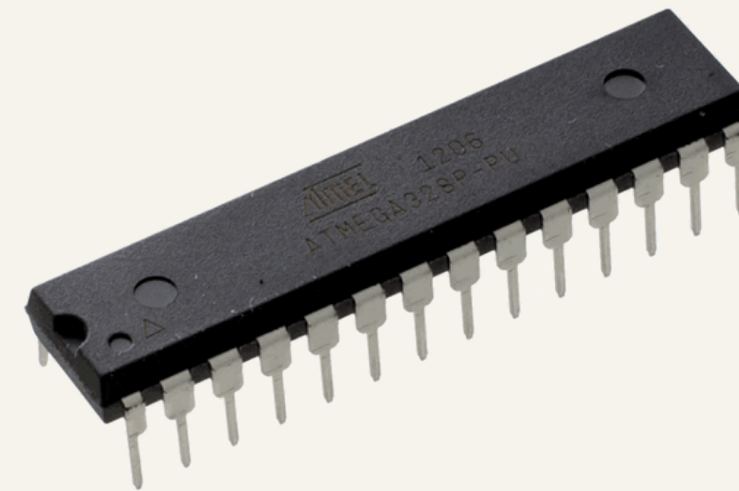
One of our team members dug up some of his old Rubik's cubes

EQUIPMENT

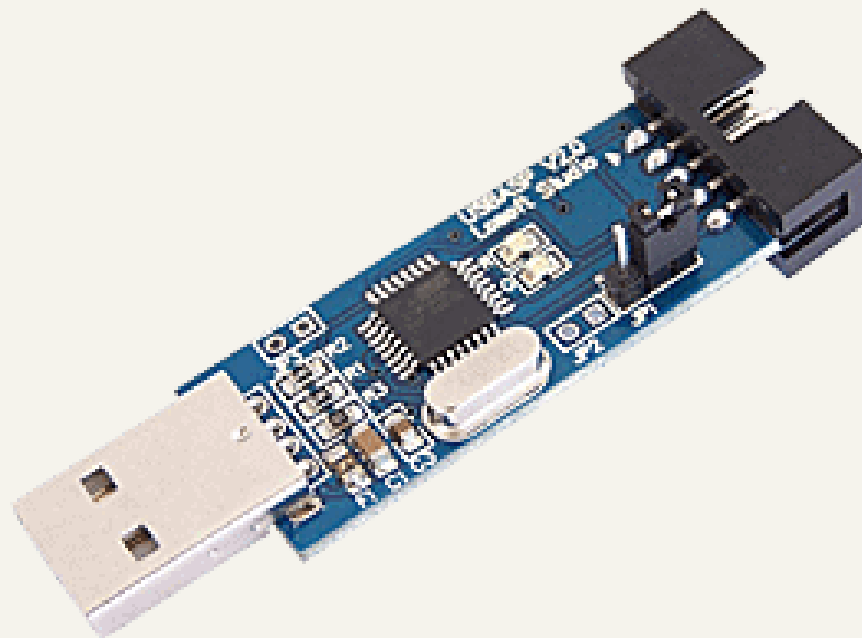
Hardware List



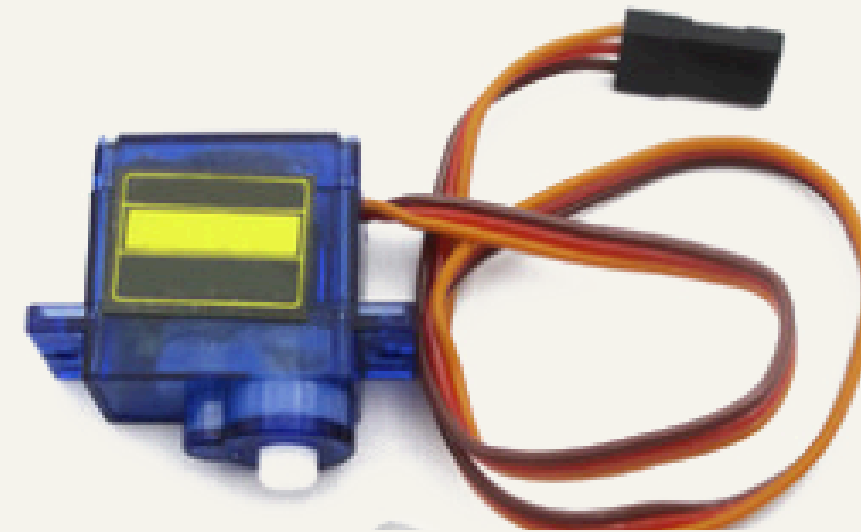
1
Rubik's Cube



2
ATMega32



3
USBASP

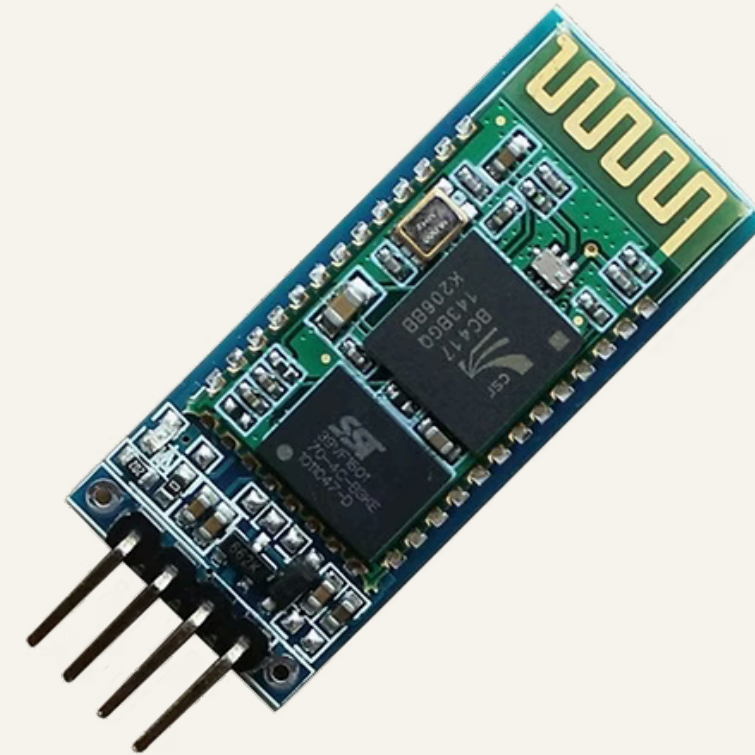


4
SG90 Servo Motor

Hardware List



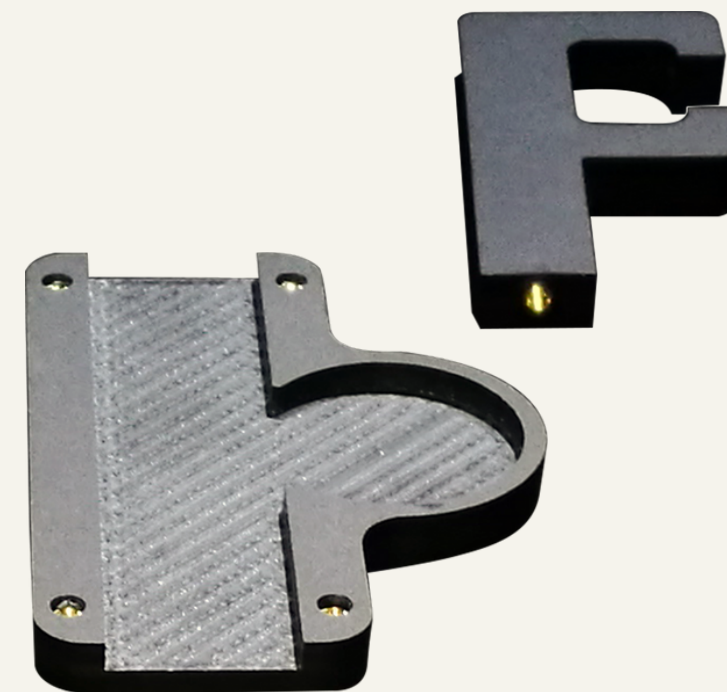
5
LCD 16x2 Display



6
HC-06 Bluetooth Module



7
Power Supply



8
Acrylic Printed
Mechanical Arm

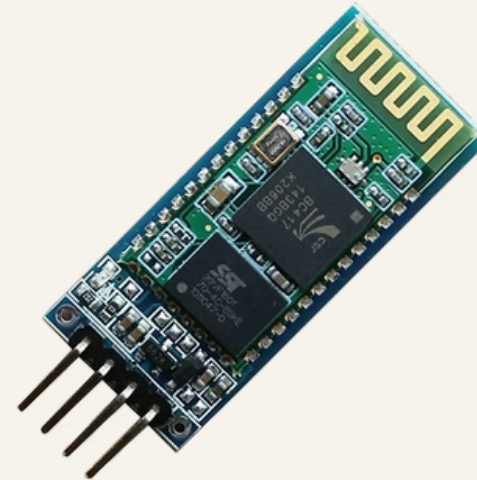
SCHEMATIC

HIGH LEVEL DIAGRAM

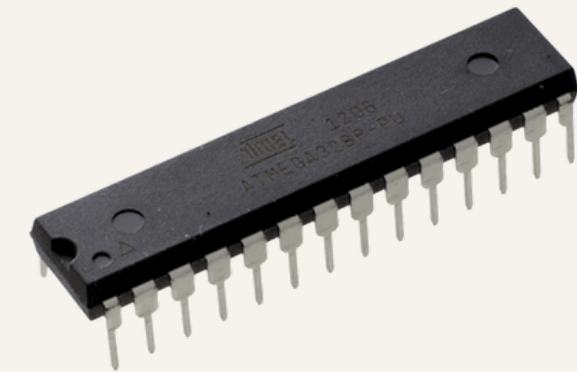
INPUT MODULE



Android app



HC-06 Bluetooth Module



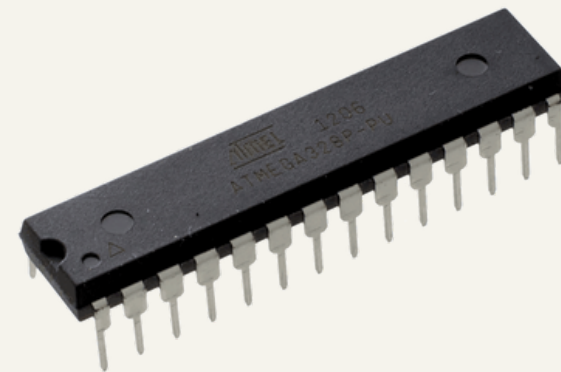
ATMega32 Microcontroller

HIGH LEVEL DIAGRAM

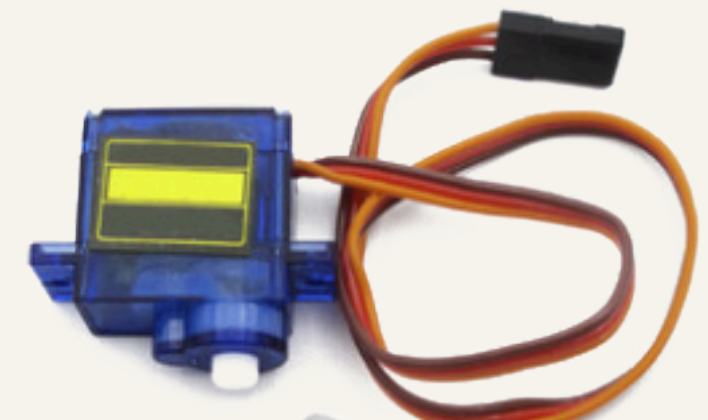
PROCESSING AND OUTPUT MODULE



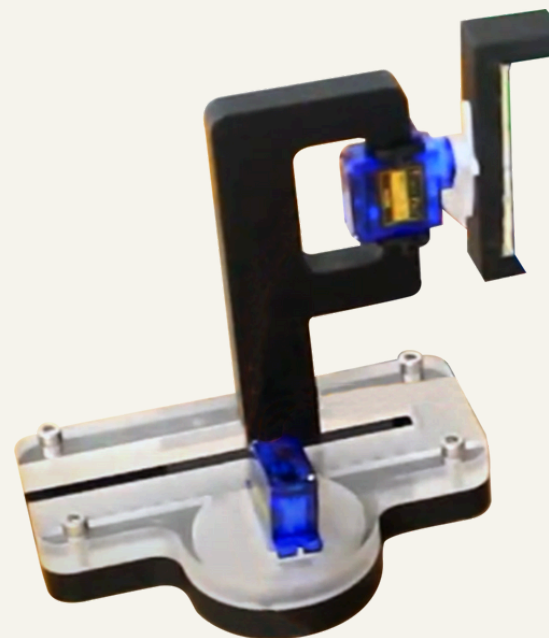
Live Solve Timer



ATMega32 Microcontroller



SG90 Servo Motor



Mechanical Arm

Explanation

- **Android App:** Captures the initial state of the Rubik's Cube using the phone camera and sends the color configuration to the microcontroller via Bluetooth.
- **HC-06 Bluetooth Module:** Acts as a wireless communication bridge between the Android app and the ATmega32 microcontroller using UART.
- **ATmega32 Microcontroller:** The core of the system that receives input from the Bluetooth module, calculates the solution (or controls motors based on a received solve sequence), and drives the mechanical components.

Explanation

- **LCD Timer Display:** Displays the time taken to complete the solve and updates live during operation.
- **Servo/Stepper Motors:** Physically manipulate the Rubik's Cube to perform the solving moves.
- **Power Supply:** Provides appropriate voltage and current to all electronic components and motors.
- **Mechanical Arm (Acrylic Structure):** Holds and rotates the Rubik's Cube based on motor control, built from custom-cut acrylic sheets.

PROCESS

Expected Outcome

The final project is expected to successfully scan the state of a standard 3×3 Rubik's Cube, compute the optimal or near-optimal solution using a solving algorithm, and physically solve the cube using motors controlled by an ATMega32 microcontroller.

Step-by-Step Plan

1. Mechanical Design

- Task: Build a frame with motors to rotate cube faces.
- Approach: Use servos/steppers; design stable holder.
- Depends on: Motor specs and structural integrity.

2. Motor Control

- Task: Drive motors via ATMega32.
- Approach: Use GPIO/PWM with motor drivers; program in C.
- Depends on: Working hardware setup.

Step-by-Step Plan

3. State Detection

- Task: Scan and read cube colors.
- Approach: Use color sensors or camera; map to data model.
- Depends on: Ability to rotate/scan all faces.

4. Solving Algorithm

- Task: Implement cube-solving logic.
- Approach: Use simplified method suitable for ATmega32.
- Depends on: Correct cube state input.

Step-by-Step Plan

5. Move Execution

- Task: Convert solution to motor actions.
- Approach: Map algorithm steps to physical moves.
- Depends on: Accurate control and valid move sequence.

6. Testing & Optimization

- Task: Ensure reliability and speed.
- Approach: Calibrate sensors/motors, debug code.
- Depends on: Full system integration.

CHALLENGES AND FUTURE DIRECTIONS

Challenges

1. Motor Control and Precision
2. Building and Assembling the Mechanical Parts
3. Cube State Detection
4. Processing Limitations of ATmega32

Future Scope

Scalability and Optimization

- The project could be enhanced to support faster solve times through optimized algorithms or by upgrading to a more powerful microcontroller while maintaining the ATmega32 for simpler tasks.

Future Scope

Integrating Advanced Visualization Systems and UI

- Adding a small display or mobile app interface could allow users to interact with the system, view solving steps, or select different solving modes

Future Scope

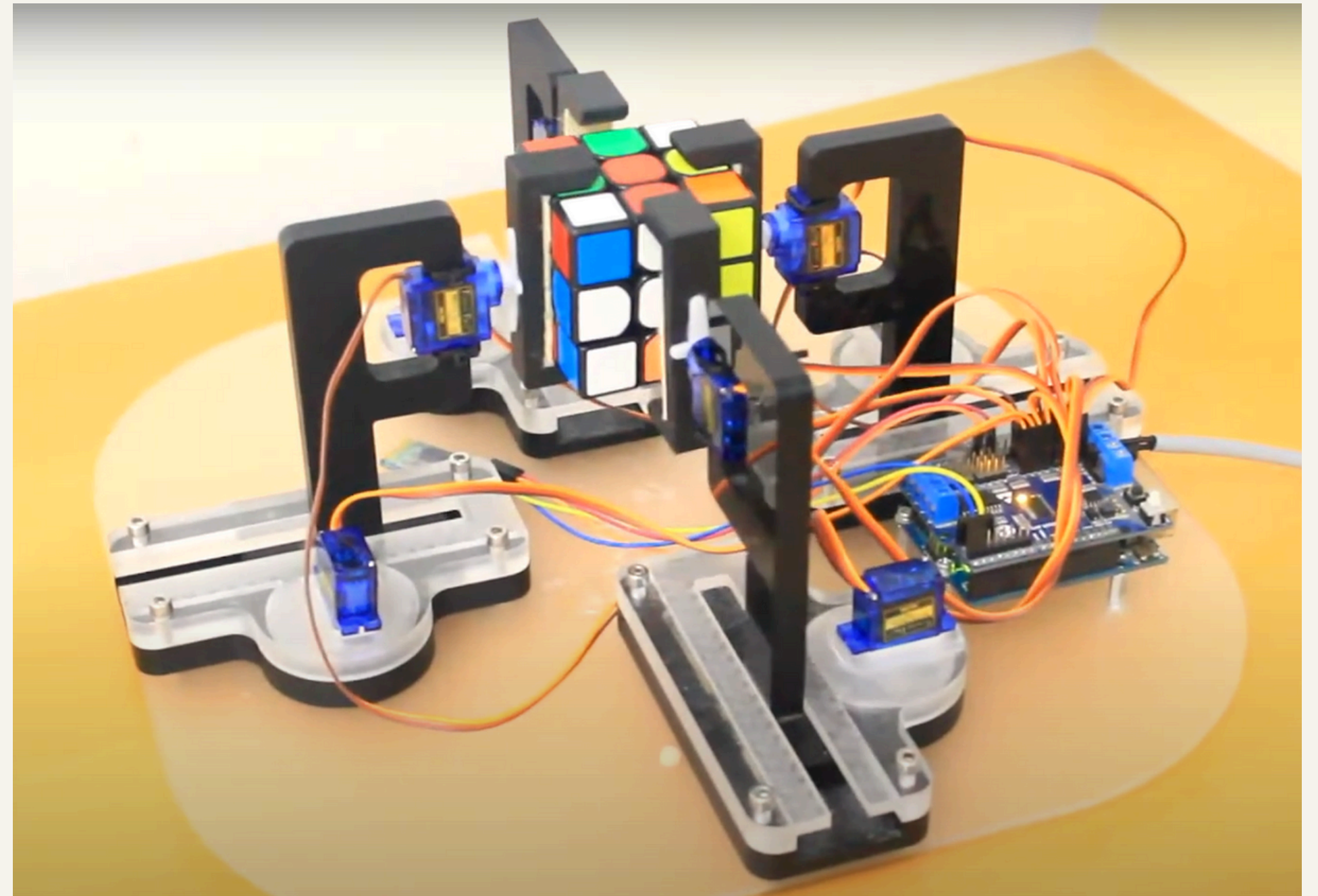
Real-World Applications Beyond the Classroom

- **STEM Demonstration:** Can be used in robotics clubs, science fairs, or tech exhibitions to demonstrate automation and embedded system applications.
- **Educational Tool:** Helps in teaching control systems, embedded programming, and algorithms.

Inspiration From

Links to the-

- [YouTube video](#)
- [Github Repository](#)



THANK YOU!